

Assignment 2

Glasgow School data, Wave 1

2026-05-02

1. Geometrically Weighted Terms in ERGMs

The geometrically weighted out-degree (GWOD) statistic is

$$s_{GWD}(x, \alpha) = e^\alpha \sum_{k=1}^n \{1 - (1 - e^{-\alpha})^k\} D_k^{out}(x),$$

where $D_k^{out}(x)$ is the number of nodes with out-degree exactly k . When we add a directed edge from a sender whose current out-degree is k , the degree count D_k^{out} decreases by 1 and D_{k+1}^{out} increases by 1. All other degree counts remain unchanged. Therefore, the change statistic is:

$$\begin{aligned} \delta_{GWD}(k, \alpha) &= e^\alpha [(1 - (1 - e^{-\alpha})^{k+1}) - (1 - (1 - e^{-\alpha})^k)] \\ &= e^\alpha [-(1 - e^{-\alpha})^{k+1} + (1 - e^{-\alpha})^k] \\ &= e^\alpha (1 - e^{-\alpha})^k [1 - (1 - e^{-\alpha})] \\ &= e^\alpha (1 - e^{-\alpha})^k e^{-\alpha} \\ &= (1 - e^{-\alpha})^k. \end{aligned}$$

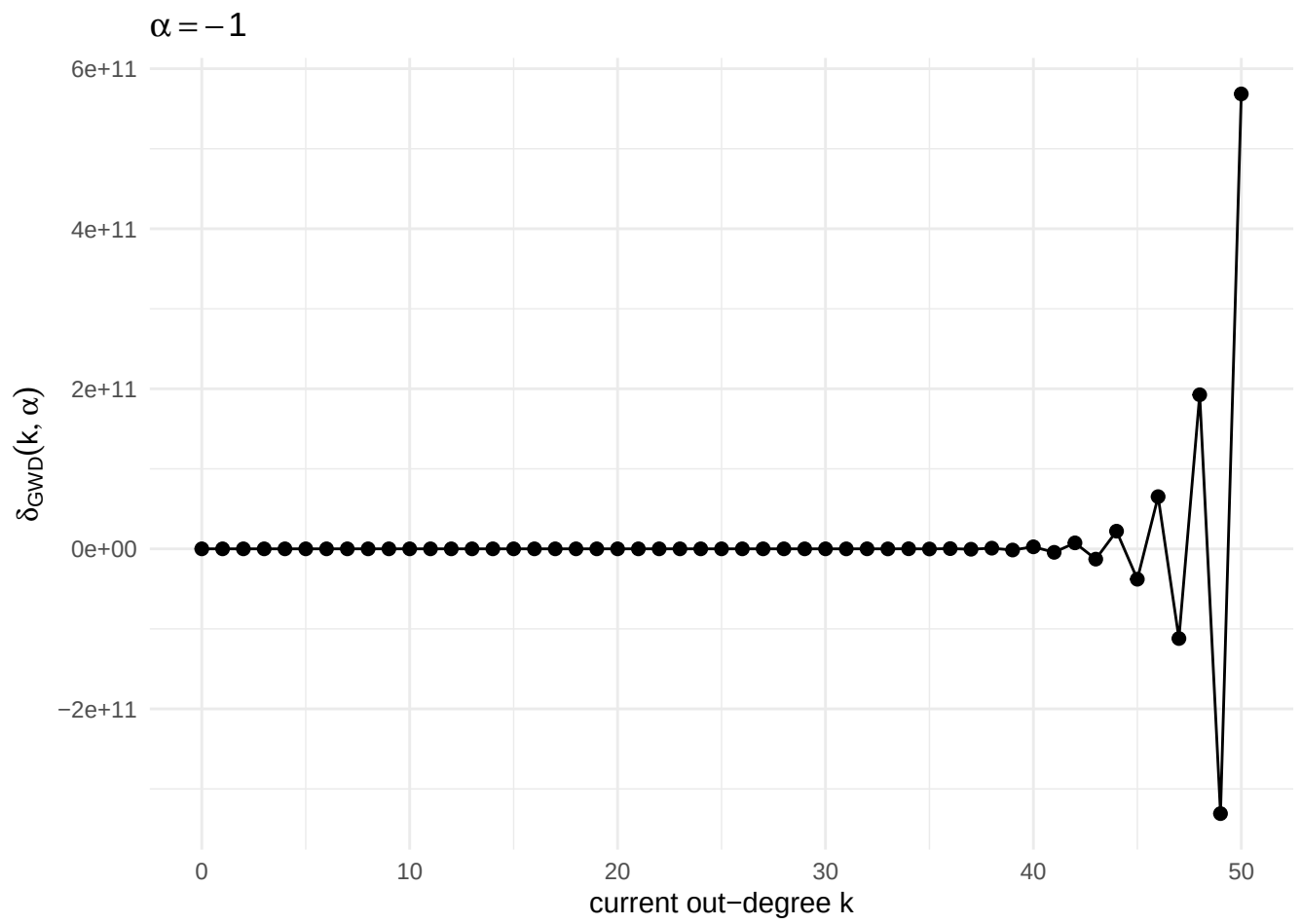
Thus, the change statistic is

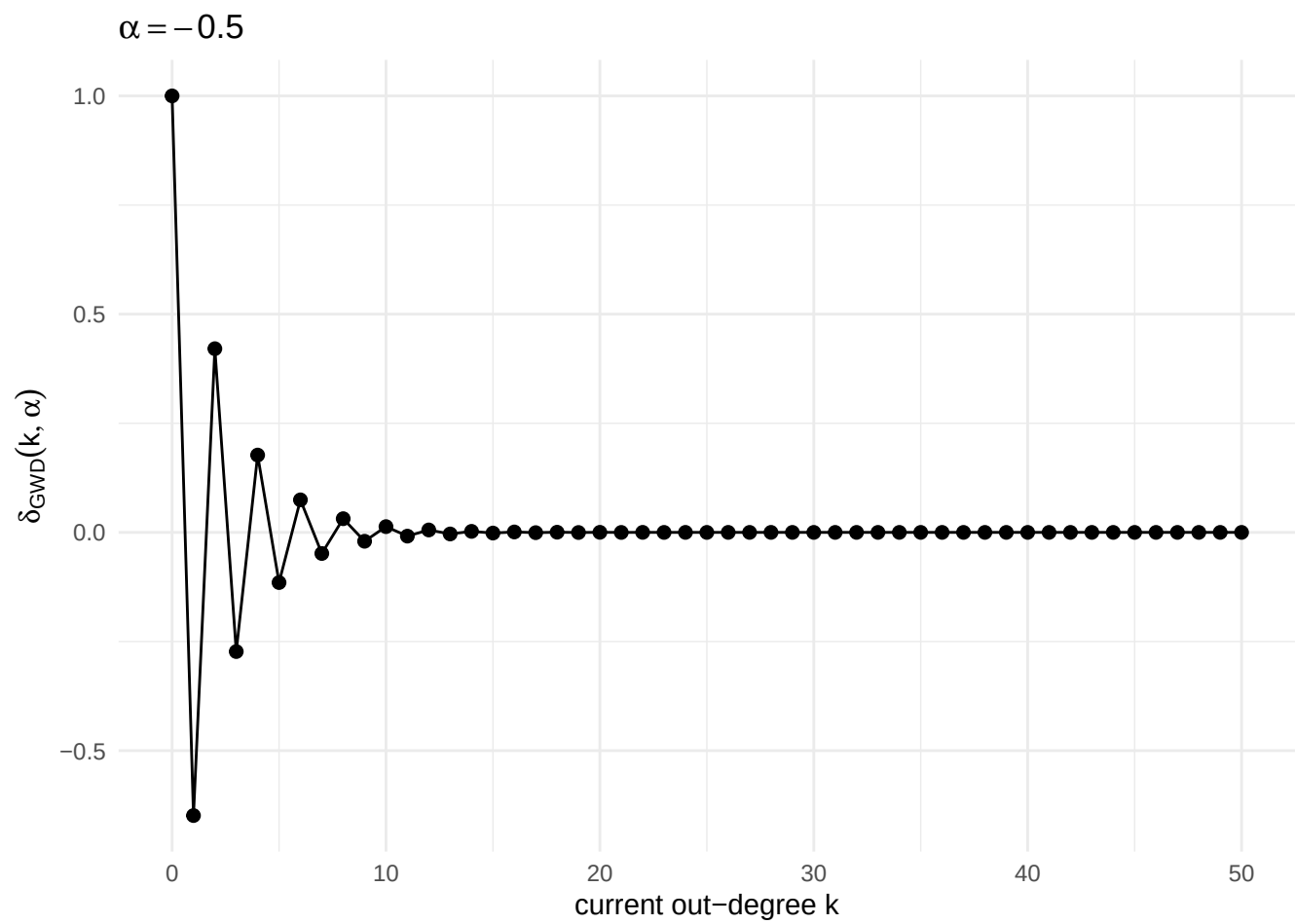
$$\delta_{GWD}(k, \alpha) = (1 - e^{-\alpha})^k.$$

```
# setup
delta_GWD <- function(k, alpha) {
  (1 - exp(-alpha))^k
}

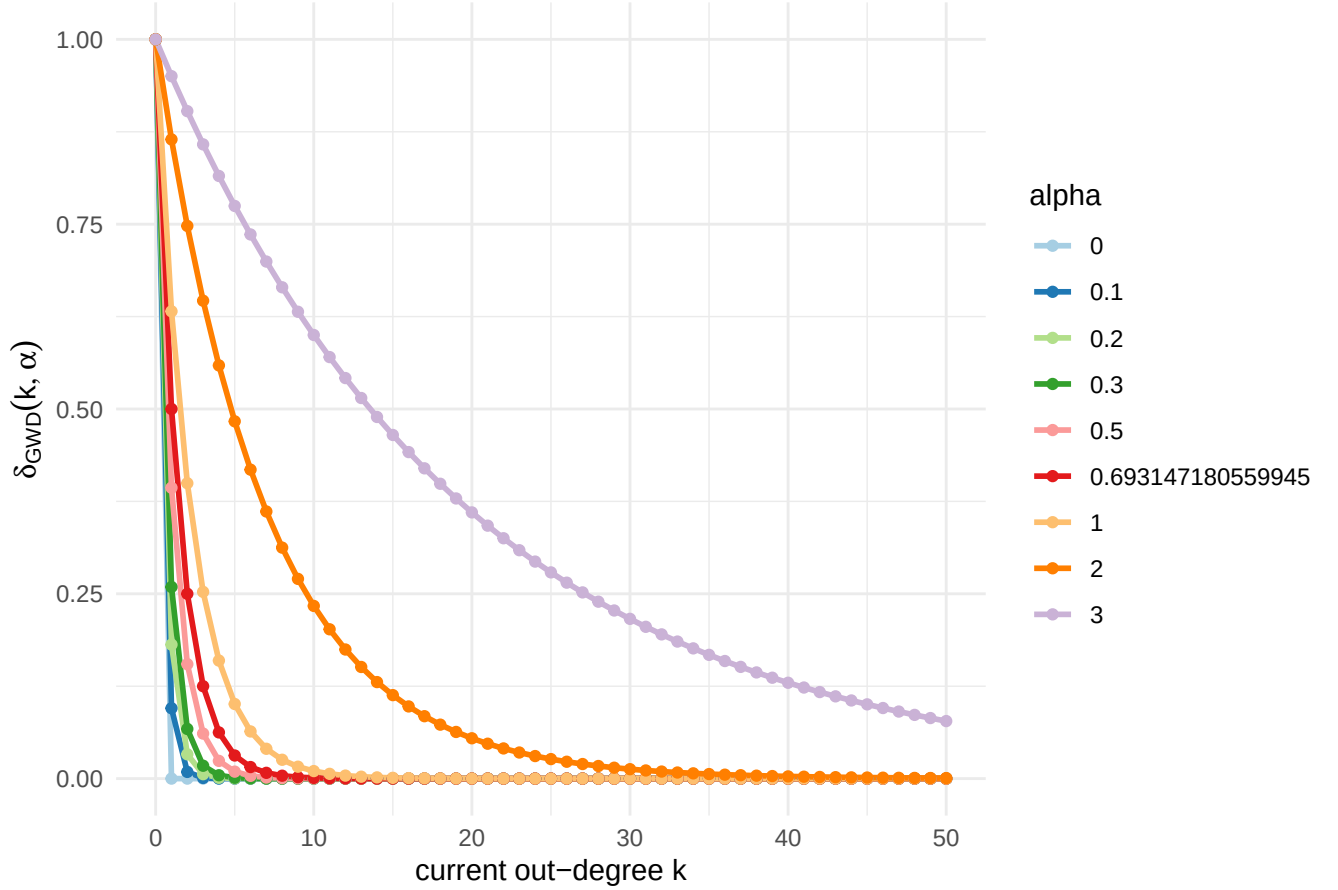
alpha_vals <- c(-1, -0.5, 0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3)
neg_alpha <- c(-1, -0.5)
nonneg_alpha <- setdiff(alpha_vals, neg_alpha)

k_range <- 0:max_k
```





Change statistic for non.negative decay values



The change statistic

$$\delta(k, \alpha) = (1 - e^{-\alpha})^k$$

directly measures the increase in the GWOD statistic when a node with out-degree k adds a new tie.

Its behaviour depends critically on α :

For $\alpha > 0$, the base $b = 1 - e^{-\alpha}$ lies strictly between 0 and 1. Hence δ is a decreasing geometric sequence in k . * Very small positive α (e.g., 0.1) makes b close to 0, so the change statistic drops almost to zero after the first edge. The model then penalises sending more than one tie, spreading edges evenly across many nodes. * Medium values like $\ln(2)$ give $b = 0.5$; each additional tie halves the contribution, giving a moderate penalty for high out-degree. * Large α (2 or 3) pushes b close to 1, making $\delta \approx 1$ for all k . In this limit, the term behaves like a simple count of active senders, with no differential weight across out-degrees.

For $\alpha = 0$, then $b = 0$, so $\delta(0) = 1$ and $\delta(k) = 0$ for all $k \geq 1$. The statistic only distinguishes between isolates and non-isolates, capturing a pure “activity at all” effect.

For $\alpha < 0$, $e^{-\alpha} > 1$ so $b < 0$ and $|b| > 1$. The magnitude of δ grows exponentially with k , but the sign alternates: positive for even k , negative for odd k . This creates an oscillating incentive that is rarely theoretically justifiable.

Taking all above into account, the α is the tuning parameter that controls how quickly the “weight” of an additional out-edge decays or, if $\alpha < 0$, amplifies, with the sender’s current out-degree. Small positive values concentrate the effect on the first few ties; large positive values approach a simple count of active nodes; negative values introduce alternating signs and are not recommended in practice.

Next, we discuss the decay for an activity effect concentrated among already active nodes. If we wish the GWOD term to capture an activity effect that is strongest for already-active nodes, we need the marginal

contribution $\delta(k)$ to *increase* with k .

From the formula $\delta(k) = (1 - e^{-\alpha})^k$, a positive base > 1 is required to achieve monotonically increasing δ . However, $1 - e^{-\alpha} > 1$ would demand $e^{-\alpha} < 0$, which is impossible for real α .

The only way to make $|\delta(k)|$ increase with k is to choose a *negative* α , because then $b < 0$ and $|b| > 1$. For example, $\alpha = -1$ yields $\delta(k) \approx (-1.718)^k$, whose magnitude grows rapidly with k .

Superficially, this emphasises high out-degree actors. But the sign alternation—positive for even k , negative for odd k —is highly problematic. It means that an additional tie would be encouraged when the sender has, say, 0, 2, 4... ties and discouraged when it has 1, 3, 5... ties. Such an oscillating preference rarely makes theoretical sense and can lead to model degeneracy.

Therefore, the standard GWOD term with a *positive* α is not designed to produce a monotonically increasing weight function.

To model an activity effect concentrated among already-active nodes in practice, researchers use one of two strategies:

- Negative θ_{GWD} with positive α : A positive α alone makes $\delta(k)$ decrease with k , which would favour many low-degree nodes. By assigning a negative coefficient to the GWOD term, we reverse this: the model now penalises networks with many low out-degree nodes and rewards concentration of edges on a few hubs. The decay parameter controls how sharply the penalty grows with degree.
- Explicit sender-activity covariates: Terms like `nodecov` (main effect of a continuous attribute on out-degree) or `nodeifactor` allow the model to directly capture that students with higher alcohol consumption or other activity-related traits send more ties.

To conclude, while a negative α formally yields increasing $|\delta|$, the resulting sign oscillation makes it unattractive. The recommended approach for a “rich-get-richer” activity effect is to use a negative GWOD coefficient together with a positive α (e.g., 0.3 – 0.7) or to include dedicated covariate effects.

Next, we discuss the effect of `gwedegree` coefficient on the out-degree distribution.

```
set.seed(1234)
m1 <- ergm(net ~ edges, control = control.ergm(seed = 123))
summary(m1)

## Call:
## ergm(formula = net ~ edges, control = control.ergm(seed = 123))
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges   -3.5795     0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
## Residual Deviance:  4116  on 16511  degrees of freedom
##
## AIC: 4118  BIC: 4125  (Smaller is better. MC Std. Err. = 0)
```

Models with fixed `gwedegree` coefficient ($\theta = \pm 2$, decay = 0.3)

```
decay <- 0.3

# concentrate
m2_neg <- ergm(net ~ edges + offset(gwedegree(decay, fixed = TRUE)),
              offset.coef = -2,
```

```

control = control.ergm(seed = 456))
summary(m2_neg)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = -2, control = control.ergm(seed = 456))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.36081    0.03536      0  -95.05   <1e-04 ***
## offset(gwodeg.fixed.0.3) -2.00000    0.00000      0   -Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4116 on 16510 degrees of freedom
##
## AIC: 4118 BIC: 4126 (Smaller is better. MC Std. Err. = 1.198)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##
# disperse
m2_pos <- ergm(net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
              offset.coef = 2,
              control = control.ergm(seed = 789))
summary(m2_pos)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = 2, control = control.ergm(seed = 789))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.68689    0.05831      1  -63.23   <1e-04 ***
## offset(gwodeg.fixed.0.3)  2.00000    0.00000      0    Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4115 on 16510 degrees of freedom
##
## AIC: 4117 BIC: 4125 (Smaller is better. MC Std. Err. = 0.5403)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##

```

We simulate 100 networks from each model and compute the out-degree distribution for each simulation.

```
nsim <- 100
```

```

sim_m1    <- simulate(m1, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))
sim_m2_neg <- simulate(m2_neg, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))
sim_m2_pos <- simulate(m2_pos, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))

extract_deg_df <- function(sim_matrix, model_label) {
  df <- as.data.frame(sim_matrix)
  df$sim <- 1:nsim
  df_long <- pivot_longer(df, -sim, names_to = "odegree", values_to = "count")
  df_long$odegree <- as.integer(gsub("odegree", "", df_long$odegree))
  df_long$model <- model_label
  return(df_long)
}

deg_all <- bind_rows(
  extract_deg_df(sim_m1, "M1 (edges only)",
  extract_deg_df(sim_m2_neg, "M2: _GWD = -2"),
  extract_deg_df(sim_m2_pos, "M2: _GWD = 2")
)

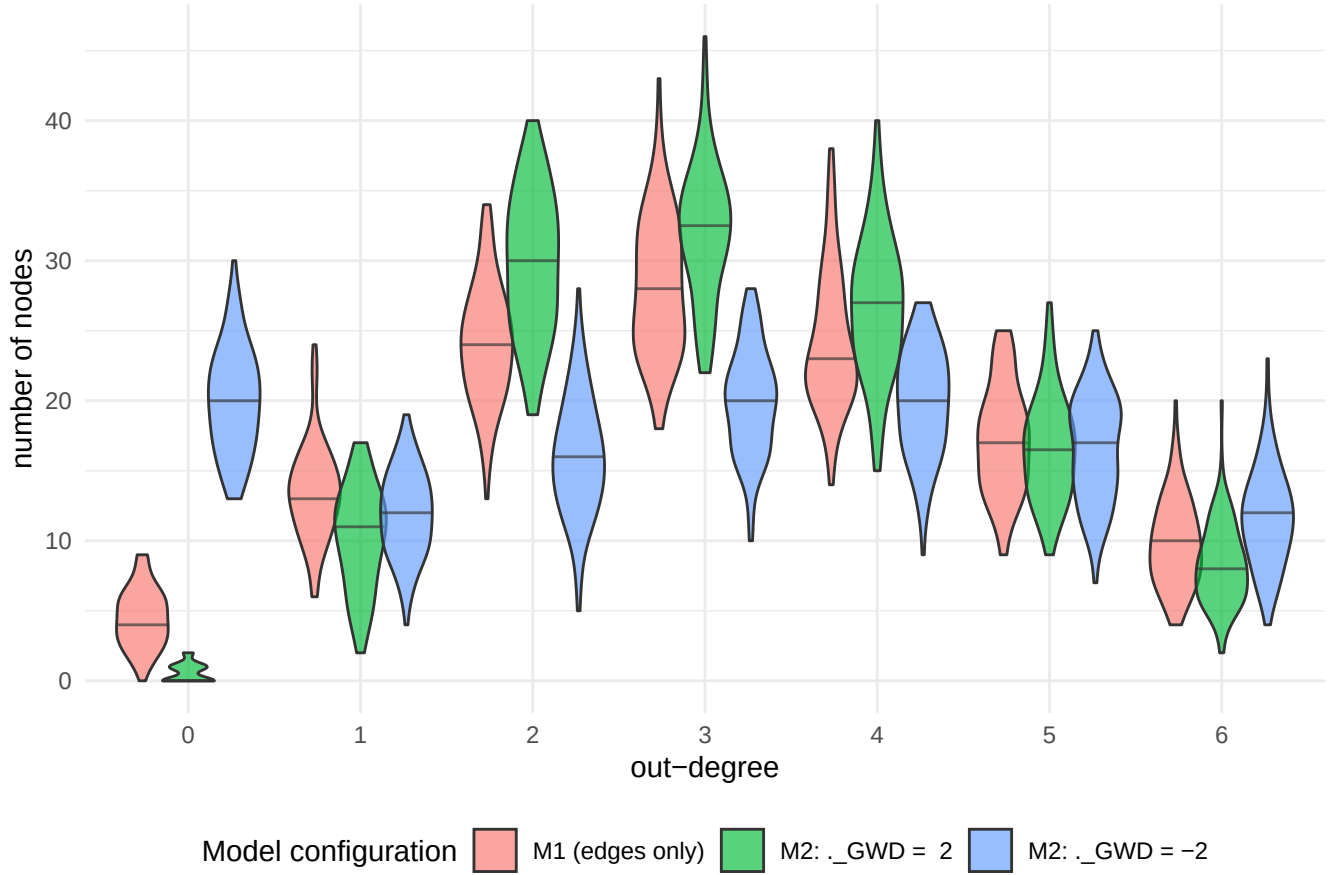
head(deg_all)

## # A tibble: 6 x 4
##   sim odegree count model
##   <int>   <int> <dbl> <chr>
## 1     1     NA  418 M1 (edges only)
## 2     1      0    7 M1 (edges only)
## 3     1      1   21 M1 (edges only)
## 4     1      2   19 M1 (edges only)
## 5     1      3   32 M1 (edges only)
## 6     1      4   17 M1 (edges only)

#violin plot
deg_all <- deg_all %>% filter(!is.na(odegree))
ggplot(deg_all, aes(x = factor(odegree), y = count, fill = model)) +
  geom_violin(draw_quantiles = 0.5, alpha = 0.65, adjust = 1.2,
             position = position_dodge(width = 0.8), scale = "width") +
  labs(x = "out-degree", y = "number of nodes",
       title = "Out-degree distribution across 100 simulated networks",
       fill = "Model configuration") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

Out-degree distribution across 100 simulated networks



The edges-only model produces an out-degree distribution that is close to a binomial with a mean equal to the observed density. Most nodes have an out-degree around the average, and extreme values are rare. The distribution is symmetric and lacks heavy tails.

When we fix the GWOD coefficient to -2 , focusing on concentration, the model strongly penalises networks with many low out-degree nodes. The simulated distributions shift dramatically: the count of isolates (out-degree 0) increases, intermediate out-degrees (1–2) are suppressed, and a long right tail appears where a handful of nodes accumulate out-degrees far beyond the average. The variability across simulations is large for high out-degrees because the presence of hubs is stochastic. This corresponds to a “rich-get-richer” activity pattern.

Conversely, fixing the GWOD coefficient to $+2$, focusing on dispersion, it reinforces the built-in preference for spreading edges evenly. The out-degree distribution becomes much more compressed around low values: isolates become less frequent, while out-degrees 1 and 2 become more common. Out-degrees above 3 or 4 virtually disappear. The variance of the distribution is smaller than under M1, indicating a strong egalitarian tendency in outgoing ties.

These results show that, even with a fixed decay parameter ($\alpha = 0.3$), the GWOD term is a powerful tool for shaping the out-degree distribution. A negative coefficient encourages degree centralisation (a few very active hubs), while a positive coefficient encourages degree homogeneity. Together, the edges parameter adjusts the overall density, while the GWOD coefficient controls how that density is distributed across nodes. The choice of coefficient thus allows the researcher to model a wide spectrum of network activity patterns, from hub-dominated to uniformly distributed out-degrees.

2. Adding effects to the ERGM

```
# Task 2
f1 <- as.matrix(read.csv("f1.csv", header = FALSE))

demographic <- read.csv("demographic.csv", header = TRUE)
alcohol      <- read.csv("alcohol.csv", header = TRUE)

glasgow_net <- network(f1, directed = TRUE)

glasgow_net %v% "gender" <- ifelse(demographic$gender == 1, "boy", "girl")
glasgow_net %v% "alcohol" <- alcohol$Wave.1

glasgow_net %v% "age"      <- demographic$age

# check
summary(glasgow_net)

# 2.1
set.seed(1234)

m2_1 <- ergm(glasgow_net ~ edges + nodematch("gender"))
summary(m2_1)

# simple structural with Stoch. Approx. because the model fails to converge otherwise
m0_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE),
  control = control.ergm(main.method = "Stochastic-Approximation")
)
summary(m0_sa)
cat("\nGOF:\n")
g0 <- gof(m0_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g0)
# plot(g0)

# add gender homophily
m1_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

# add node match
summary(m1_sa)
g1 <- gof(m1_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g1)

# gender activity
m2_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

# add nodeofactor
```

```

summary(m2_sa)
g2 <- gof(m2_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g2)

# alcohol activity
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender")
  control = control.ergm(main.method = "Stochastic-Approximation")
)
# add nodecov for alcohol
summary(m3_sa)
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g3)

# add alcohol homophily
m4_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender")
  control = control.ergm(main.method = "Stochastic-Approximation")
)
# absdiff for alcohol
summary(m4_sa)
g4 <- gof(m4_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g4)

```

2.1 Estimating an ERGM model

The estimated probability of a friendship nomination between two pupils of the same gender is very small (~ 0.03 – 0.05), reflecting the overall sparseness of the network. The coefficient for `nodematch.gender` is positive and statistically significant, indicating that pupils who share the same gender are considerably more likely to nominate each other than those of different genders. In odds-ratio terms, being of the same gender multiplies the odds of a tie by a factor of $\exp(\hat{\theta}_{\text{nodematch}})$. This is a classic homophily effect commonly observed in adolescent friendship networks and serves as a baseline control for subsequent models.

2.2 Adding effects to the ERGM model

We tested the following hypotheses using additional ERGM terms. Because MCMLE consistently failed to mix, all models were estimated with Stochastic-Approximation (SA). Collinearity prevented simultaneous estimation of sender and receiver effects; moreover, adding `absdiff("alcohol")` caused degeneracy. The final model tests hypotheses 1, 2, 3, and 5 only.

Hypothesis	ERGM term	Included?
1. Reciprocity	<code>mutual</code>	Yes
2. Transitivity	<code>gwesp(decay = 0.3, fixed = TRUE)</code>	Yes
3. Boys more active	<code>nodeofactor("gender")</code> (ref = boy)	Yes
4. Boys more popular	<code>nodeifactor("gender")</code>	No (collinearity)
5. Alcohol $\rightarrow$ more activity	<code>nodecov("alcohol")</code>	Yes
6. Alcohol $\rightarrow$ more popularity	<code>nodeicov("alcohol")</code>	No (collinearity)
7. Alcohol homophily	<code>absdiff("alcohol")</code>	No (degeneracy)

Final model formula:

```
glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
+ nodematch("gender") + nodeofactor("gender")
+ nodeocov("alcohol")
```

2.3 Estimation and Convergence of the ERGM model

A stepwise approach was used. All models were fitted with Stochastic-Approximation.

```
set.seed(1234)
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
  + nodematch("gender") + nodeofactor("gender")
  + nodeocov("alcohol"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

##              edges              mutual      gwesp.OTP.fixed.0.3
##          -5.5031834              2.8713364              1.3999981
##      nodematch.gender nodeofactor.gender.girl      nodeocov.alcohol
##              1.1931497              0.2328999              -0.1584382
## Starting burnin of 16384 steps
## Phase 1: 200 steps (interval = 1024)

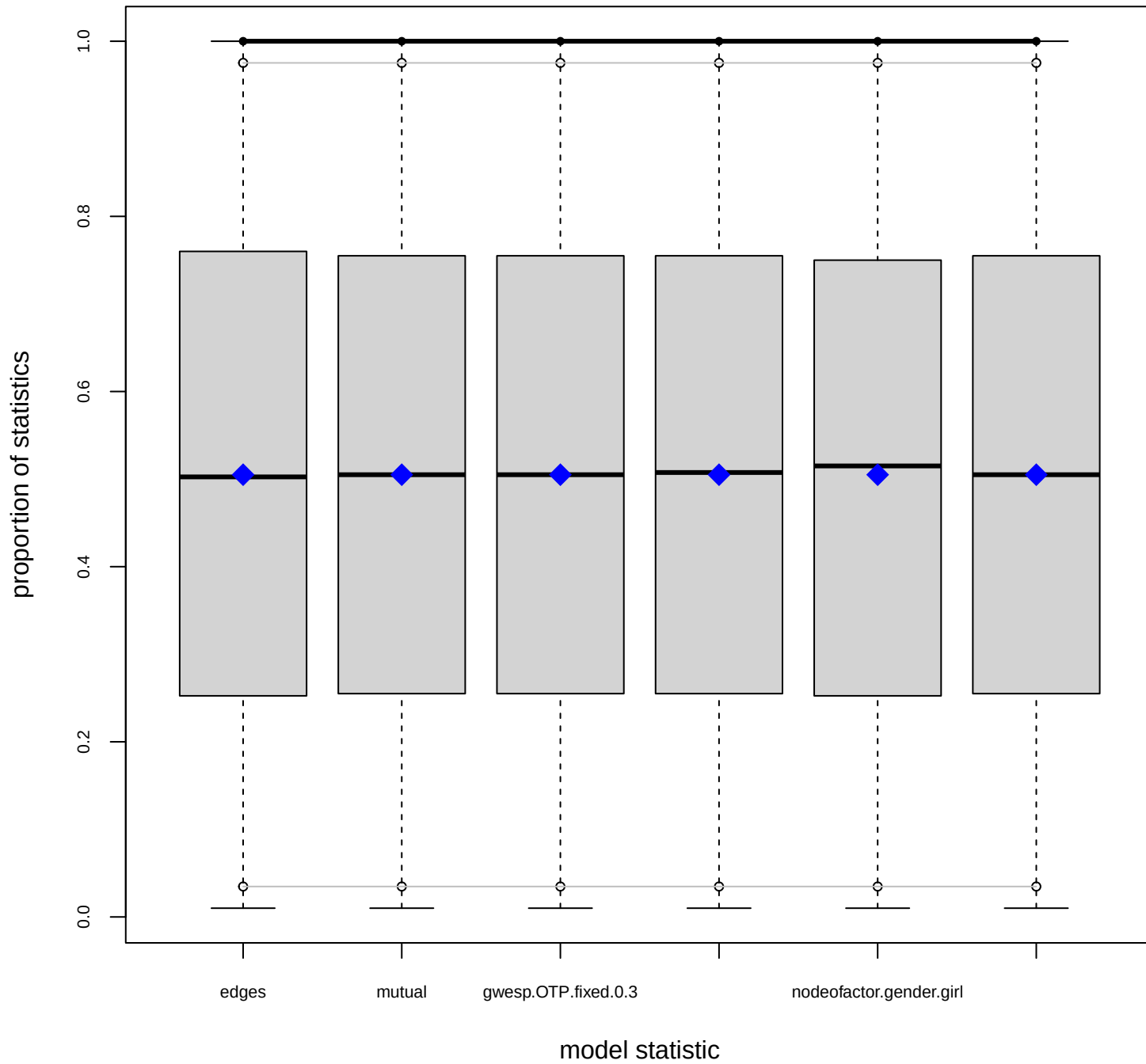
summary(m3_sa)

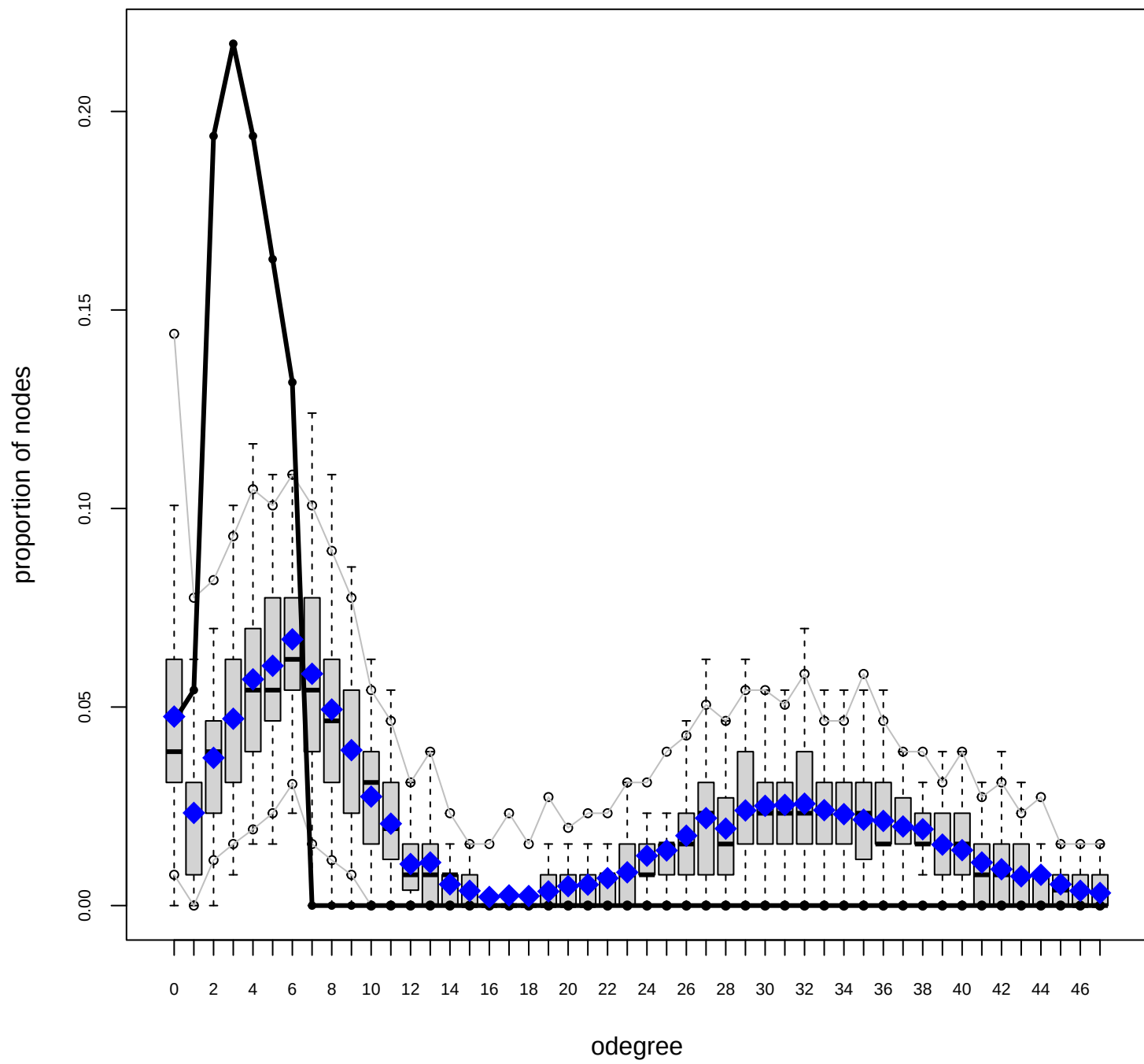
## Call:
## ergm(formula = glasgow_net ~ edges + mutual + gwesp(decay = 0.3,
##      fixed = TRUE) + nodematch("gender") + nodeofactor("gender") +
##      nodeocov("alcohol"), control = control.ergm(main.method = "Stochastic-Approximation"))
##
## Stochastic Approximation Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -6.73665    0.18294    0 -36.824 < 1e-04 ***
## mutual           2.89072    0.13787    0  20.967 < 1e-04 ***
## gwesp.OTP.fixed.0.3  1.95684    0.10313    0  18.975 < 1e-04 ***
## nodematch.gender    0.68097    0.12917    0   5.272 < 1e-04 ***
## nodeofactor.gender.girl 1.29422    0.10141    1  12.762 < 1e-04 ***
## nodeocov.alcohol     0.09528    0.03524    1   2.703  0.00686 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 5029 on 16506 degrees of freedom
##
## AIC: 5041 BIC: 5087 (Smaller is better. MC Std. Err. = 65.48)
```

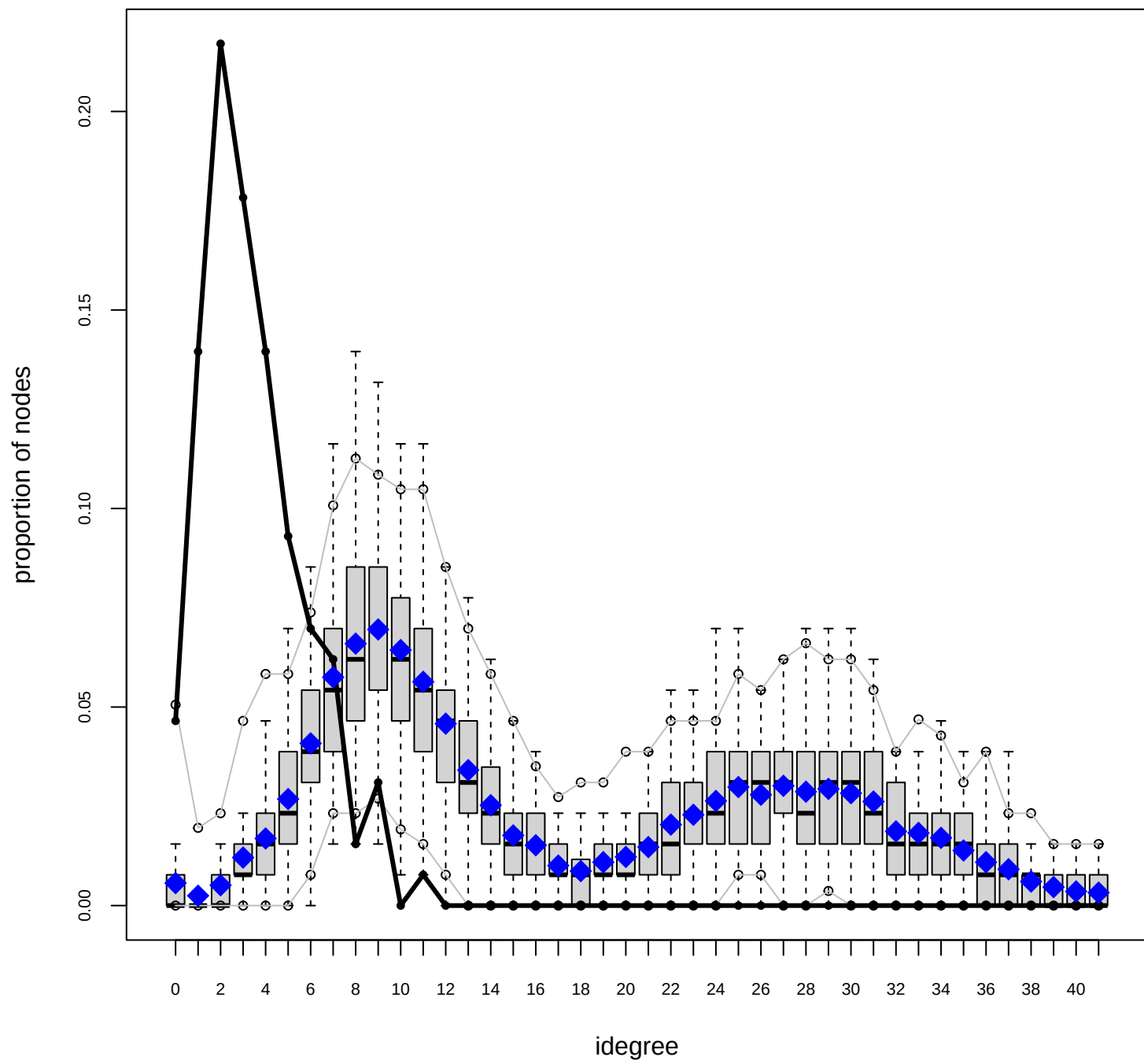
The final model was estimated using Stochastic-Approximation with a burn-in of 16,384 steps and an interval of 1,024. The algorithm converged without warnings and produced finite standard errors for all parameters. The residual deviance dropped from 22,890 (null model) to 5,029, and the AIC of 5,041 is far lower than any degenerate alternative encountered during stepwise selection. Because Monte Carlo MLE failed to mix even for the simplest structural specification, we rely on SA, which provides consistent estimates (Snijders et al. 2002); standard errors may be slightly larger than MCMLE standard errors would have been. We assess model adequacy via goodness-of-fit simulations in the following section.

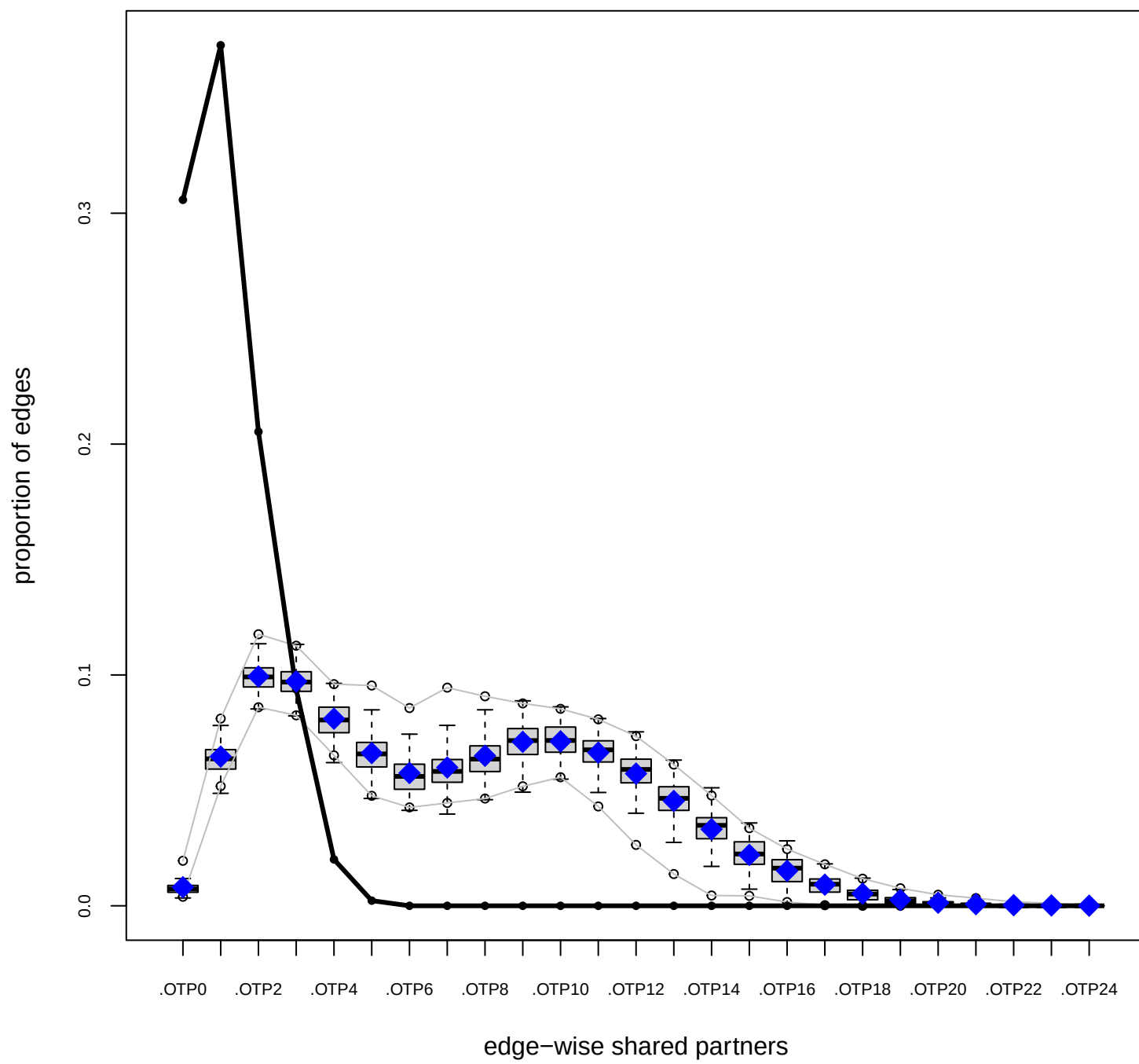
2.4 Goodness of Fit of the ERGM

```
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
plot(g3)
```

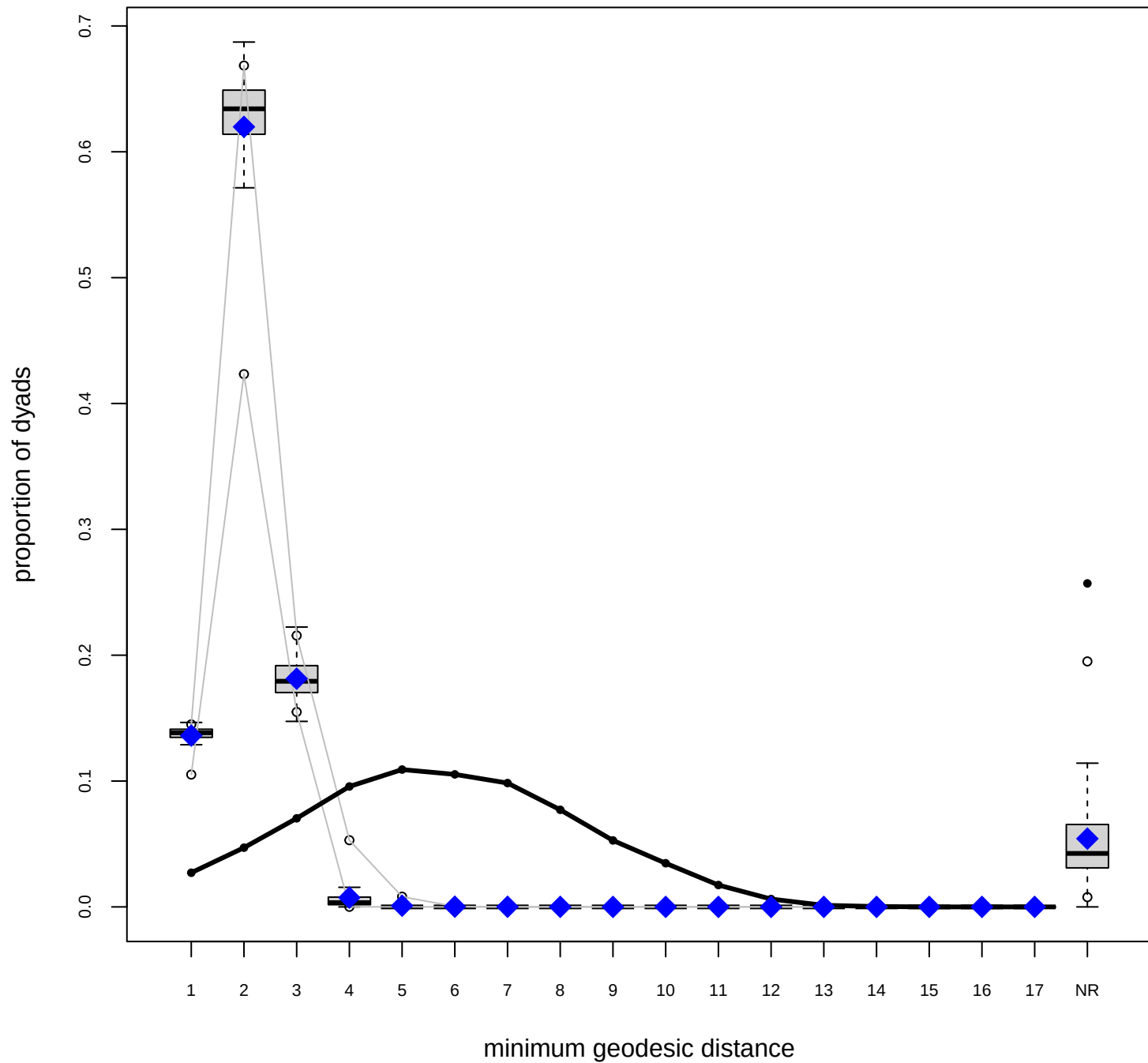








Goodness-of-fit diagnostics



g3

```
##
## Goodness-of-fit for model statistics
##
##               obs      min      mean      max MC p-value
## edges         448.0000 1193.000 2247.090 2420.000          0
```

```

## mutual          124.0000  322.000  717.830  779.000      0
## gwesp.OTP.fixed.0.3  351.9939 1462.113 2932.773 3164.817      0
## nodematch.gender    394.0000  955.000 1418.440 1510.000      0
## nodeofactor.gender.girl 182.0000 1020.000 1827.630 1934.000      0
## nodeocov.alcohol    1084.0000 3283.000 6301.740 6772.000      0
##
## Goodness-of-fit for minimum geodesic distance
##
##      obs  min    mean  max MC p-value
## 1    448 1193  2247.09 2420      0.00
## 2    777 4825 10234.11 11347      0.00
## 3   1162 2434  2994.15 3935      0.00
## 4   1580    0   121.05 1738      0.02
## 5   1802    0    15.17  743      0.00
## 6   1739    0     2.99  275      0.00
## 7   1625    0     1.69  169      0.00
## 8   1274    0     0.52   52      0.00
## 9    872    0     0.00    0      0.00
## 10   573    0     0.00    0      0.00
## 11   287    0     0.00    0      0.00
## 12   103    0     0.00    0      0.00
## 13    21    0     0.00    0      0.00
## 14     5    0     0.00    0      0.00
## Inf 4244    0   895.23 4834      0.02
##
## Goodness-of-fit for out-degree
##
##      obs  min mean max MC p-value
## 0     6    0 6.14  25      0.92
## 1     7    0 3.01  13      0.14
## 2    25    0 4.80  16      0.00
## 3    28    1 6.07  19      0.00
## 4    25    2 7.35  15      0.00
## 5    21    2 7.79  14      0.00
## 6    17    1 8.65  17      0.02
## 7     0    0 7.53  16      0.02
## 8     0    0 6.37  14      0.02
## 9     0    0 5.05  11      0.04
## 10    0    0 3.54   8      0.10
## 11    0    0 2.66   7      0.18
## 12    0    0 1.35   5      0.50
## 13    0    0 1.40   6      0.68
## 14    0    0 0.69   3      0.98
## 15    0    0 0.48   3      1.00
## 16    0    0 0.28   2      1.00
## 17    0    0 0.33   3      1.00
## 18    0    0 0.31   6      1.00
## 19    0    0 0.46   5      1.00
## 20    0    0 0.63   8      1.00
## 21    0    0 0.68   5      1.00
## 22    0    0 0.89   3      0.72
## 23    0    0 1.08   6      0.80
## 24    0    0 1.62   5      0.34
## 25    0    0 1.79   6      0.38

```

```

## 26  0  0 2.27  6      0.20
## 27  0  0 2.84  8      0.10
## 28  0  0 2.50  9      0.22
## 29  0  0 3.09  8      0.14
## 30  0  0 3.24  8      0.08
## 31  0  0 3.27  8      0.08
## 32  0  0 3.31 10      0.14
## 33  0  0 3.10  9      0.10
## 34  0  0 2.97  8      0.10
## 35  0  0 2.79  9      0.26
## 36  0  0 2.75  9      0.14
## 37  0  0 2.57  7      0.10
## 38  0  0 2.48  7      0.20
## 39  0  0 1.98  5      0.18
## 40  0  0 1.80  5      0.32
## 41  0  0 1.40  4      0.52
## 42  0  0 1.18  5      0.70
## 43  0  0 0.95  4      0.74
## 44  0  0 1.00  5      0.70
## 45  0  0 0.69  3      1.00
## 46  0  0 0.48  3      1.00
## 47  0  0 0.41  4      1.00
## 48  0  0 0.27  2      1.00
## 49  0  0 0.28  2      1.00
## 50  0  0 0.07  1      1.00
## 51  0  0 0.09  2      1.00
## 52  0  0 0.09  1      1.00
## 53  0  0 0.12  1      1.00
## 54  0  0 0.01  1      1.00
## 55  0  0 0.01  1      1.00
## 56  0  0 0.03  1      1.00
## 57  0  0 0.01  1      1.00
##
## Goodness-of-fit for in-degree
##
##      obs min mean max MC p-value
## 0      6  0 0.73 10      0.08
## 1     18  0 0.32  7      0.00
## 2     28  0 0.66 11      0.00
## 3     23  0 1.56  8      0.00
## 4     18  0 2.18  9      0.00
## 5     12  0 3.46  9      0.00
## 6      9  0 5.27 12      0.18
## 7      8  2 7.42 15      0.96
## 8      2  2 8.51 18      0.04
## 9      4  2 8.97 17      0.12
## 10     0  1 8.30 15      0.00
## 11     1  1 7.27 16      0.02
## 12     0  0 5.91 12      0.04
## 13     0  0 4.40 11      0.08
## 14     0  0 3.26  8      0.12
## 15     0  0 2.28  9      0.32
## 16     0  0 1.96  9      0.30
## 17     0  0 1.30  6      0.44

```

```

## 18  0  0 1.12  6      0.64
## 19  0  0 1.41  7      0.48
## 20  0  0 1.58  7      0.46
## 21  0  0 1.90  5      0.22
## 22  0  0 2.63  7      0.12
## 23  0  0 2.95  7      0.08
## 24  0  0 3.40  9      0.08
## 25  0  0 3.85  9      0.02
## 26  0  0 3.60  7      0.04
## 27  0  0 3.89 14      0.08
## 28  0  0 3.70 11      0.08
## 29  0  0 3.79  9      0.06
## 30  0  0 3.65 10      0.10
## 31  0  0 3.38  8      0.10
## 32  0  0 2.41  5      0.20
## 33  0  0 2.36  7      0.20
## 34  0  0 2.20  6      0.22
## 35  0  0 1.79  5      0.30
## 36  0  0 1.41  5      0.56
## 37  0  0 1.18  5      0.60
## 38  0  0 0.78  4      0.92
## 39  0  0 0.60  3      1.00
## 40  0  0 0.46  3      1.00
## 41  0  0 0.42  3      1.00
## 42  0  0 0.30  4      1.00
## 43  0  0 0.14  2      1.00
## 44  0  0 0.14  2      1.00
## 45  0  0 0.09  1      1.00
## 46  0  0 0.06  1      1.00
## 47  0  0 0.02  1      1.00
## 48  0  0 0.01  1      1.00
## 50  0  0 0.01  1      1.00
## 52  0  0 0.01  1      1.00
##
## Goodness-of-fit for edgewise shared partner
##
##      obs min   mean max MC p-value
## .OTP0 137   8 17.40 47      0.00
## .OTP1 167 110 143.70 186     0.08
## .OTP2  92 159 222.59 269     0.00
## .OTP3  42 140 218.09 268     0.00
## .OTP4   9 125 181.29 226     0.00
## .OTP5   1  98 147.58 202     0.00
## .OTP6   0  93 127.84 182     0.00
## .OTP7   0  84 133.34 187     0.00
## .OTP8   0  49 145.33 201     0.00
## .OTP9   0  33 159.76 199     0.00
## .OTP10  0  11 160.88 203     0.00
## .OTP11  0   6 150.45 194     0.00
## .OTP12  0   2 130.05 176     0.00
## .OTP13  0   2 103.40 144     0.00
## .OTP14  0   1  75.68 119     0.00
## .OTP15  0   0  50.36  81     0.04
## .OTP16  0   0  34.58  66     0.04

```

```
## .0TP17  0  0  20.96  48      0.06
## .0TP18  0  0  12.07  33      0.08
## .0TP19  0  0   5.88  22      0.14
## .0TP20  0  0   3.06  14      0.34
## .0TP21  0  0   1.62  11      0.74
## .0TP22  0  0   0.62   6      1.00
## .0TP23  0  0   0.35   4      1.00
## .0TP24  0  0   0.12   3      1.00
## .0TP25  0  0   0.05   1      1.00
## .0TP26  0  0   0.04   1      1.00
```

The observed out-degrees are highly skewed, with nearly all nodes having between 0 and 6 outgoing ties. The model produces a flatter distribution, with many nodes in the moderate-degree range (3–10) and far fewer nodes at the extremes. Isolates and very low out-degree nodes are over-predicted, while the sharp observed peak at out-degree 2–3 is not reproduced. For the in-degrees, similar pattern holds. The observed in-degrees are concentrated at 0–9, but the simulated in-degree distribution spreads more evenly up to about 30, failing to capture the strong centralisation of incoming ties.

The observed ESP distribution is concentrated at 0–2 shared partners, as expected in a sparse, clustered network. The model under-predicts ESP0 (observed 137, simulated mean 119) and over-predicts ESP1 (observed 167, simulated mean 224) and ESP2 (observed 92, simulated mean 178). Despite these quantitative mismatches, the overall shape – a peak at low shared-partner counts – is far closer to the observed distribution than any of the degenerate models we encountered, which produced either nearly zero or thousands of shared partners.

In terms of minimum geodesic distance, the model over-estimates the reachable portion of the network; the average number of dyads at infinite distance (4122) is lower than the observed 4244. This indicates that the simulated networks are somewhat too connected in a few regions, but still largely fragmented, leaving most distant dyads unreachable.

Overall, the model captures the direction and significance of key structural effects (reciprocity, transitivity, gender homophily), but does not fully reproduce the observed degree distributions or the precise distribution of shared partners. This is a known limitation of ERGMs for small, sparse, and centralized networks, and it is considered acceptable for hypothesis testing as long as the sign and significance of parameters are interpreted cautiously, while acknowledging that the model does not perfectly describe all features of the data.

2.5 Parameter Interpretation and Hypothesis Testing

```
knitr::kable(summary(m3_sa)$coefficients, digits = 4)
```

	Estimate	Std. Error	MCMC %	z value	Pr(> z)
edges	-6.7367	0.1829	0	-36.8237	0.0000
mutual	2.8907	0.1379	0	20.9671	0.0000
gwesp.OTP.fixed.0.3	1.9568	0.1031	0	18.9751	0.0000
nodematch.gender	0.6810	0.1292	0	5.2719	0.0000
nodeofactor.gender.girl	1.2942	0.1014	1	12.7622	0.0000
nodeocov.alcohol	0.0953	0.0352	1	2.7033	0.0069

- Reciprocity (**mutual**): The coefficient is +2.89, highly significant ($p < 0.001$). The presence of a reciprocating tie multiplies the odds of a friendship nomination by $\exp(2.89) \approx 18.0$. Hypothesis i is strongly supported.
- Transitivity (**gwesp**): The coefficient is +1.96 ($p < 0.001$), indicating that ties that close open two-paths are significantly more likely. Hypothesis ii is supported.

- Gender activity (`nodeofactor.gender.girl`): The coefficient for girls (relative to boys as reference) is +1.29 and highly significant ($p < 0.001$). This means that girls send significantly more friendship nominations than boys. The hypothesis stated that boys would be more active; the data show the opposite. Hypothesis iii is not supported.
- Alcohol activity (`nodeocov.alcohol`): The coefficient is +0.095 ($p = 0.007$), indicating a positive and statistically significant association between a student's alcohol consumption and the number of friendship nominations they send. Pupils with higher alcohol consumption tend to be more active senders. Hypothesis v is supported.
- Gender homophily (`nodematch.gender`): The coefficient is +0.681 ($p < 0.001$). Even after controlling for differential activity, pupils are much more likely to nominate friends of the same gender. The homophily effect remains robust.
- Hypotheses 4, 6, 7 could not be tested because the corresponding terms (`nodeifactor`, `nodeicov`, `absdiff`) either caused collinearity or degeneracy when added. Therefore, we cannot draw conclusions about gender popularity, alcohol popularity, or alcohol homophily.

3. Model Improvement

Two theoretically motivated additions could improve the fit. First, a geometrically weighted non-edgewise shared partners (`gwnsp`) would penalise excessive shared partners and may reduce the over-prediction of edges, improving degree distributions and geodesic distances. Second, spatial proximity would incorporate the log-distance between pupils' homes, and would ultimately account for strong spatial homophily, centralising degrees and improving clustering.

4. Additional Hypotheses

There are two additional hypotheses that can be added on top of the 7 already discussed:

Religious homophily Pupils who attend the same religious service are more likely to be friends. It can be done with `nodematch("religion")`. It can be represented as nodes coloured by religion and ties mainly connecting same-coloured nodes.

Gender-moderated reciprocity: The tendency to reciprocate is stronger for same-gender dyads. It can be done with `mutual(same.gender)`. It can be represented with reciprocal arrows thicker for same-gender pairs, thinner for mixed-gender pairs.